

L21 Vol 16 Substrate Algebra Chapter Scaffolding

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 15:55 EDT

L21 Vol 16 Substrate Algebra Chapter Scaffolding v0.1

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track E L21: Vol 16 Substrate Algebra chapter-by-chapter scaffolding per L19 v0.3 outline (12-chapter structure).

1. Vol 16 12-Chapter Scaffolding

Ch 1: A_{sub} Operator Algebra Foundations

- Definition of A_{sub} as substrate-natural operator algebra
- Substrate operations: substrate-Bergman + substrate-Pochhammer + substrate-Casimir
- Foundational substrate-natural axioms
- Cross-link to substrate-K-type spectral theory

Ch 2: Substrate K-Types as Algebra Modules

- Substrate-K-type $V_{(\lambda_1, \lambda_2)}$ per $SO(5) \times SO(2)$ representation theory
- Substrate-K-type spinor tower $V_{(1/2, 1/2)}, V_{(3/2, 1/2)}, V_{(5/2, 1/2)}$
- Substrate-K-type integer-weight series $V_{(1, 0)}, V_{(1, 1)}, V_{(2, 0)}$
- Substrate-K-type Sym^k decomposition per $V_{\text{fund}} \text{Sp}(2)$

Ch 3: $A_{\text{sub}} \leftrightarrow$ Hall Algebra Dictionary

- Hall algebra structure constants per Q_{B2} substrate-quiver
- Green coproduct + Cartan grading substrate-natural form
- Engine at operator level (per Vol 16 §3 + §4 v0.1)
- Substrate-process dictionary per 6 SM processes engine-verified

Ch 4: Engine at Operator Level

- Engine v0.1 Green coproduct + process-dictionary
- Engine v0.2 canonical/crystal basis of $U_{q^+}(B_2)$
- Engine v0.3 R-matrix / scattering S-matrix
- Engine v0.4 negative part / antimatter (Drinfeld double F-generators)

Ch 5: Substrate-Bergman Kernel Algebra (NEW Thursday)

- Substrate-Bergman kernel $K_B(z, w) = c_{FK} \cdot (1 - z \cdot \bar{w})^{\{-n_C/\text{rank}\}}$
- Saturday May 28 One-Genus convention $n_C/\text{rank} = 5/2$ substrate-canonical
- $c_{FK} \cdot \pi^{(9/2)} = 225 = (N_c \cdot n_C)^2$ substrate-natural (Sunday G5.1 PASS)
- Substrate-Bergman matrix element at substrate-K-type $V_{(\lambda_1, \lambda_2)}$ per Helgason Ch. IX

Ch 6: Schur II Pochhammer at Substrate-K-Types (NEW Thursday)

- Substrate-Pochhammer $(n_C/\text{rank})_k$ per $\text{Sym}^k V_{\text{fund}} \text{Sp}(2)$
- Pochhammer cascade per spinor tower $(5/2)_1, (5/2)_3, (5/2)_5$
- Substrate-Pochhammer $\times$ substrate-Casimir composite per substrate-K-type
- Substrate-Pochhammer Schur II rule substrate-canonical derivation

Ch 7: Substrate-Symplectic Cat 6 via $\text{Sp}(2) \cong \text{Spin}(5)$ (NEW Thursday)

- $\text{Sp}(2) \cong \text{Spin}(5)$ accidental Lie-group isomorphism
- Substrate-symplectic 2-form ω substrate-canonical antisymmetric
- Substrate-Hamiltonian flow at substrate-K-type spectrum
- Substrate-time evolution $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{\text{BST}})$ substrate-canonical

Ch 8: Substrate-Schur Generators Registry (NEW Thursday)

- 15 catalogued SSG + 6 candidate (SSG-16 to SSG-21)
- Cat A independent count 5-7 effective per Cal #235 + Keeper SSG Audit v0.3
- A/B/C classification per SSG Registry v0.22
- 6 substrate-mechanism FORCING categories

Ch 9: Composite Substrate-Mass-Mechanism at gen-2 (NEW Thursday)

- Composite v0.4-v0.5 FORWARD derivation per L4-5a-e
- Substrate-Bergman + substrate-Pochhammer Schur II + substrate-deficit per Casey #13
- Tier 1/Tier 2 boundary precision 0.112%
- Multi-week substrate-mass-mechanism FORCING cascade

Ch 10: Casey #14 STANDING Substrate-Predicted 3+1 Minkowski (NEW Thursday)

- Substrate-codim-4 substrate-mechanism cascade
- $1/n_C$ chirality projection per substrate-Shilov boundary
- Substrate-Predicted 3+1 Minkowski signature substrate-natural
- Multi-week RIGOROUS-tier promotion via Strong-Uniqueness 11 STANDALONE legs

Ch 11: Cal #36 STANDING Positive-Search Operational (NEW Thursday)

- 7-element discipline stack CLOSED operational maturity
- Schur lemma operationalization as audit discipline
- One-Primitive-Many-Observables substrate-PMNS Cat A primitive cluster
- 8-brake Lyra-on-self brake-cascade Thursday afternoon operational

Ch 12: Substrate Hall Algebra Paper Cross-Reference

- Cross-reference Paper L8 v0.3 Substrate Hall Algebra primary paper
- Multi-week v0.4 continuation plan per Cal #189
- Cross-link to all Vol 16 chapters

2. Multi-Week Vol 16 v0.4 Continuation

substantive multi-week: - substantive chapter-by-chapter content extension per Cal #189 - substantive cross-CI Keeper K-audit per chapter - substantive Cal cold-read substantive per chapter

3. Closure v0.1

L21 Vol 16 Substrate Algebra chapter scaffolding v0.1: substantive 12-chapter scaffolding per L19 v0.3 outline + Thursday substrate-mechanism cascade absorbed.

substantive multi-week Vol 16 v0.4 continuation per Cal #189 + Keeper K-audit + Cal cold-read.

Tier: L21 Vol 16 chapter scaffolding v0.1 framework; substantive multi-week per Cal #189.

— Lyra, Thu 2026-06-04 11:56 EDT. L21 Vol 16 Substrate Algebra chapter scaffolding v0.1: 12-chapter scaffolding per L19 v0.3 outline + Thursday substrate-mechanism cascade absorbed (Ch 5-11 NEW Thursday content); multi-week v0.4 per Cal #189.